DESIGN AND IMPLEMENTATION OF LEAD-ACID BATTERY STATE-OF-HEALTH AND STATE-OF-CHARGE MEASUREMENTS

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Abstract – **In many electrical and electronic applications there is the need to have an efficient measurement over the State-of-Health (SoH) and the State-of-Charge (SoC). The SoH is an indirect measurement about the age of the battery while the SoC indicates the energy stored in the battery. The SoC and SoH are commonly referred in papers and scientific reports. However, their design and implementation procedure is very often left behind. Moreover, the implementation is not trivial. This paper presents the design and implementation of lead-acid battery SoH and SoC. The purpose is to present carefully the procedure showing how to design and implement the SoH and SoC estimators. The SoH estimation is based on obtaining the battery internal resistance. The SoC is based on the Coulomb Counting method, valid for charging and discharging processes. Experimental results using a 12V/2.3Ah battery show the efficacy of the proposed design and implementation method.**

Keywords – **Coulomb Counting, Internal-Resistance, Lead-Acid Battery, Ohm's Law Application, State-of-Charge, State-of-Health.**

I. INTRODUCTION

The search for alternative ways to replace the generation of electric energy that comes from the use of nonrenewable sources is very important for the current situation of the electric sector. Seeking to generate energy through renewable sources, various devices capable of converting other types of energy into electric energy have been developed. For example, solar panels that transform solar energy into electric energy or wind turbines to transform kinetic energy from wind into electric energy. These devices need to store the generated energy because wind and sunshine are not always available. Additionally, these sources of energy vary in different timescales, from some seconds to hours and days [1]. In this way, storing energy becomes indispensable to overcome the intermittency behavior of non-renewable sources.

A way to store electrical energy is through the use of batteries. A battery is a set of cells connected in series. These cells can be manufactured in various combinations of materials. Some examples of materials used for the construction of these cells are nickel-cadmium (NiCd), Lithium-ion (Li-ion), Lithium polymer (Li-ion polymer), lead-acid (Pb) and Sodium-Sulphur (NaS) [2]–[4].

The choice of what battery technology best fits an application is based on several criteria like robustness, cost, life-cycle, power density, temperature, depth of discharge and so on [5], [6].

Lithium-ion batteries are the primary choice for portable electronics applications and they are very attractive for electric vehicles due to as efficiency around 95%, long life cycle (3000 cycles) and high energy and power densities. However, their relative high cost and the need for careful circuitry for safety and protection are their main hurdles [7].

Nickel-cadmium batteries are attractive for Uninterrupted Power Supply (UPS) and generator-starting applications. However, the sales of these batteries declined for the period of 1995 to 2003, motivated mainly by the increasing environmental controls for toxic cadmium.

Sodium-Sulphur batteries are the most promising technology for variable renewable energy sources, such as wind and solar power. In the past decades, installations based on sodium-Sulphur grew exponentially from 10 MW in 1998 to 305 MW at the end of 2008. Sodium-Sulphur batteries can be cycled 2500 times, have high power density, have general capacity around 1MW and have efficiency around 75 – 90%. Additionally, they are environmentally benign and 99% of the overall weight of the battery material can be recycled. The drawbacks of sodium-Sulphur batteries are the high capital cost and the necessity to keep the internal temperature around 350 °C [8].

Lead-acid batteries have been used for more than 130 years. Lead acid batteries have high reliability, low cost, strong surge capability, efficiency around 65 to 80 %. Even though lead-acid batteries are being replaced in a variety of scenario for lithium-ion batteries, they still have preferable in some niche of applications. Lead-acid batteries are attractive for UPS systems, power quality devices and for spinning reverse applications [3].

When batteries are used in projects, it is usually essential to know how much energy is stored and available to be used. This information can be obtained through the SoC estimation. Another valuable information about batteries is the SoH. With these information, it is possible to have a greater control over the energy that is stored and therefore facilitating the energy management of the entire system.

Estimators for SoC and SoH of lead-acid battery have been constantly presented in the literature [9]–[14]. In [9]–[11], the authors proposed a SoC estimator based on Kalman filtering method. The estimators were verified with simulation and experimentally. A fuzzy SoC estimator was proposed in [12] for lead-acid batteries used in electric vehicles. The proposed estimation can avoid undercharging or overcharging at the batteries, avoiding a reduction in the battery life-cycle. Concerning SoH estimators, in [13] the authors presented a SoH prediction based on artificial neural

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network model. The results showed accuracy and convergence of the model operating in a laboratory condition. In [14], the authors presented an estimation strategy based on adaptive control theory for online parameters identification. To speed up the estimator's convergence, the adaptation law was replaced by a genetic algorithm. All these estimators have their efficacy and validity. However, new issues can be addressed.

Focusing on presenting a simple but efficient way to measure the SoC and SoH of lead-acid batteries, this article aims to demonstrate how to design and implement an estimator for lead-acid batteries using methods of Coulomb Counting to estimate the SoC and evaluating the internal resistance with the help of Ohm's law to estimate the SoH. Experimental results show the efficacy of the proposed methods.

II. DESIGN

As previously stated, currently there are many types of batteries in the market, each one with its advantages and disadvantages. One of the longest in the market is the lead-acid battery. These have as advantages the large temperature range in which they can be operated. One disadvantage of lead-acid batteries is the low number of cycles, staying between 400 and 2000 cycles.

Despite the new batteries that are emerging, such as lithium batteries and lithium polymer batteries, there is still plenty of room in the market for lead acid batteries, especially in systems where its temperature is not controlled and robustness is required. With this in mind, the choice to work with lead-acid batteries to construct the estimator of the SoC and SoH is justified.

A. State of Charge (SoC)

The SoC of a battery is a percentage value that represents the amount of energy that is stored at the time it is being observed. One of the most commonly used methods to estimate the SoC value is called the Coulomb Counting [15]–[17]. This method basically controls the amount of charge that is stored inside the battery through (1):

$$SoC = \frac{Ca - \int idt}{Ca} \times 100\% \tag{1}$$

where C_a is the discharge capacity of the battery, that is, the amount of energy the battery can store in a way that does not compromise its state of health. The energy that is withdrawn from the battery is measured by the integral of the current at time t. Equation (1) gives us the percentage of energy that is stored in a battery.

B. State of Health (SoH)

To estimate the SoH of a lead-acid battery, (2) is used [17], [18].

$$SoH = \frac{R_{EoL} - R_i}{R_{EoL} - R_N} \times 100\% \tag{2}$$

Where R_{EoL} is the internal resistance of the battery when it is in a state considered as the end of life, that is, the capacity to store charge is considerably below when a battery is new. This value in general is informed by battery manufacturer. The R_n is the internal resistance of the battery when it is new. This value is also informed by the battery manufacturer. And R_i is the actual internal resistance of the battery. To find this value, one can use methods such as evaluating internal resistance through the use of Ohm's law. In order to use Ohm's law it is necessary to apply a pulse to the battery. After the pulse is applied, it will return information about the voltage and current, thus, using Ohm's law, it is possible to measure the actual internal resistance of the same (3).

$$R_i = \frac{V_B - V_{B0}}{I_B - I_{B0}} \tag{3}$$

Where V_B being the voltage, and I_B being the battery current in response to the pulse and V_{BO} and I_{BO} are the actual voltage and current of the battery, respectively.

III. IMPLEMENTATION

This section presents the implementation of the SoC and SoH estimators. All procedures are carefully described as well as how to assemble a prototype for experimental validation of the estimators.

A. General devices

To implement the prototype, it is necessary to use some general devices, as follows:

1) Sensors

To estimate the SoC and the SoH it is necessary to have a real-time monitoring of the battery. In this way, through the use of voltage sensors and current sensors, it is possible to have information about the present behavior of the battery. For the prototype, two current sensors LA 55-P (LEM Transducer Current) and one voltage sensor LV 55-P (LEM Voltage Transducer) were used.

2) Microcontroller:

With the data generated by the sensors, after being conditioned, it is necessary to process this data to provide information about SoC and SoH. To process this information the Texas Instruments TMS320F28335 processor module was used.

3) Solid-state Relay

The solid-state relay Ssr-25dd is used as a switch to apply a pulse to the battery. To switch on the relay, the microcontroller sends a pulse, which in turn, passes through a buffer circuit.

B. Signal Conditioning

Having the data generated by the sensors, it is possible to process them using the processor, thus obtaining the estimates on SoC and SoH. However, to use the information of the current sensors and the voltage sensor, it is necessary to perform the conditioning of these signals in order to reduce their amplitude, but keeping the information intact, thus allowing the processor to process the signals from the sensors correctly.

For the signal conditioning, the circuit of Figure 1 was used. This circuit, through the use of operational amplifiers and potentiometers, is able to apply a gain to the input signal, thus changing the amplitude of the signal but maintaining the signal information. For the prototype, three signal conditioning circuits were used, one for each sensor, each circuit was adjusted to apply a gain of 1/10 in each signal.

C. State of Health (SoH):

In order to measure the health status of the battery, it is necessary some battery information, i.e. the internal resistance of the battery when the battery is new and the internal resistance of the battery at the end of life. These informations can be obtained from the manufacturer.

The actual internal resistance is obtained using the strategy of applying a pulse to the battery and using the pulse response information that the battery emits. To generate the pulse, an external, adjustable source connected to the battery was used, according to Figure 2. A solid-state relay is used as switch S1.

The voltage and current sensors are being measured and the response of the battery to the pulse is observed. The theoretical battery response curves are shown in Figure 3(a) and Figure 3(b), where the curves in Figure 3(a) is the actual current curve (dashed line), ideal current curve (continuous line), and the curve in Figure 3(b) is be the curve of voltage.

To estimate the actual internal resistance of the battery, the Ohm law, is used (3), with values referring to the difference ΔU ($V_B - V_{B0}$) for the voltage and ΔI ($I_B - I_{B0}$) for the current, as shown in Figure 3(a) and Figure 3(b).

Thus, with all values obtained (R_{EoL}, R_N e R_i), it is enough to apply (2) to obtain the state of health of the battery in percentage.

The SoH is better expressed in percentage. Thus, its value indicates the current state of health of the battery. Table I presents the SoH classification according to the percentage of SoH.

D. Obtaining the discharge capacity (C_a).

The discharge capacity is a parameter usually supplied by the battery manufacturer. However, the C_a value changes from battery to battery, mainly due to their internal physical differences. Therefore, it is necessary to obtain the discharge capacity of the battery being used.

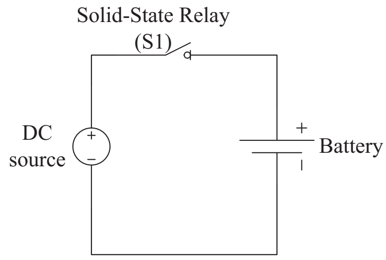

Fig. 2. Simplified diagram for connecting an adjustable source to the battery.

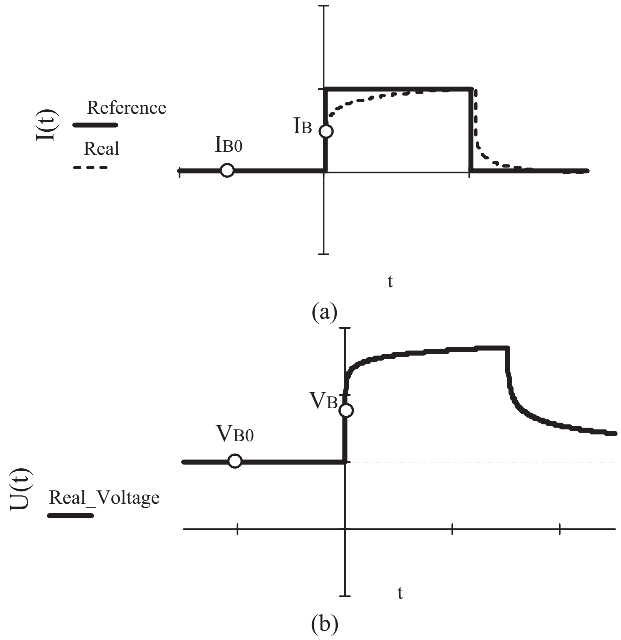

Fig. 3. (a) Theoretical battery current response to pulse and (b) theoretical battery voltage response to pulse. V_{B0} is the actual voltage of the battery.

TABLE I
SoH Classification

Classification	Percentage Range of Values
Excellent	$95\% < SoH \leq 100\%$
Good	$90\% < SoH \leq 95\%$
Regular	$85\% < SoH \leq 90\%$
Bad	$80\% < SoH \leq 85\%$
Very Bad	$SoH \leq 80\%$

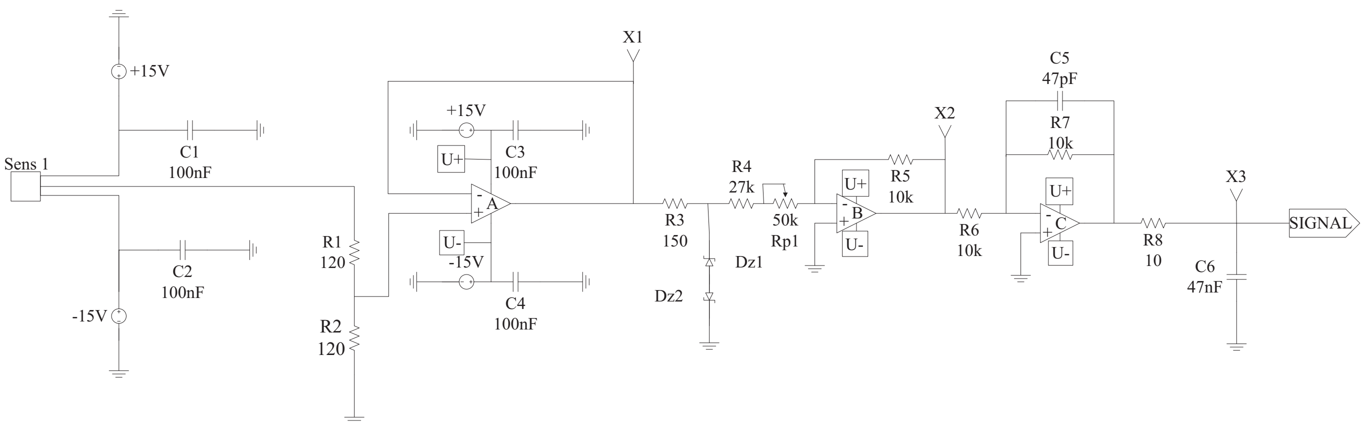

Fig. 1. Signal Conditioning Circuit

To obtain the value of C_a a method is used by evaluating the amount of charge the battery uses when it is discharged from the fully charged point to the point where it is considered totally discharged.

The fully charged point of the battery can be considered when the voltage at the battery terminals is 1.5V above the rated voltage of the battery.

The point where the battery is considered fully discharged is obtained while the battery is being discharged. The observed curve in the discharge of a battery has an almost constant behavior of supplying power from the beginning of the discharge to a point where there is an abrupt drop in the power supplied. The point where the battery is considered to be fully discharged should be chosen based on the value at which this abrupt drop occurs.

In addition to the points considered as the fully charged and fully discharged battery, a value for C_a must be stipulated. The choice of this value should take into account the battery charge capacity reported by the manufacturer in Ampère-Hour (Ah). Since it is only a value to be used to obtain the effective value of C_a, this first value must be oversized, and a value should be considered for each unit of Ah. In this paper, this value is five thousand.

With the values set, the battery is discharged using a simple resistive circuit as shown in Figure 4. The battery discharge must be monitored to observe the moment when the sudden drop in the supplied power begins to occur. In addition to this measurement, it is necessary to perform the SoC measurement algorithm, with the stipulated value of C_a.

To find the effective C_a value, it is calculated using the SoC value found and the C_a value stipulated above (4).

$$Ca = \frac{(100 - SoC) \times Ca_{sti}}{100} \tag{4}$$

Where C_{asti} is the stipulated value and SoC is the value obtained with the battery discharge.

Thus, therefore, the discharge capacity of the evaluated battery is being found.

With the value of C_a properly obtained, it is enough to follow the current by current sensor and using (1) to obtain the value of SoC in percentage.

E. Code to estimate SoC and SoH:

Equations (1) and (2) are implemented in order to obtain SoC and SoH measurements. Figure 5 presents the algorithm flowchart used to estimate SoC. If the current sensor is in the positive direction, i.e. the battery supplying the load, the integral of the battery current is discounted from C_a, respecting (1). However, if the current is negative, i.e the battery is being charged, the integral of the battery current is increased by C_a. However, if the current is zero, so that neither of the current sensors detects current flow, the SoC value must remain constant to show that there is no charge or discharge to the battery.

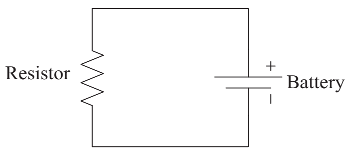

Fig. 4. Resistive circuit for battery discharge.

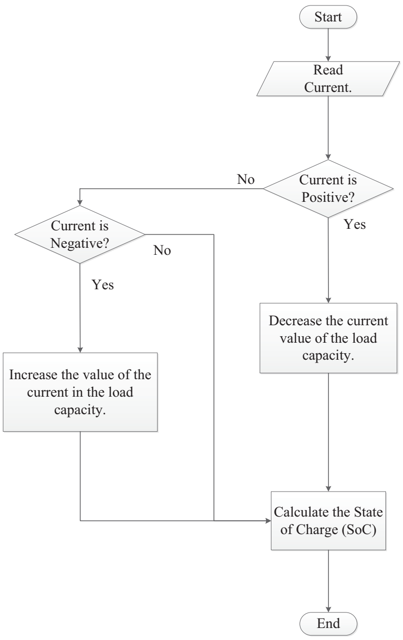

Fig. 5. Flowchart to measure the SoC of the battery.

In order to obtain the current internal resistance of the battery (R_i), the flowchart of Figure 6 was designed. Since the intent of this algorithm is to capture the points of the curve in response to the pulse sent to the battery, the flowchart indicates that if the current sensor signal and the voltage sensor signal increase for forty samples, the response to the pulse was captured. This allows the reading of the internal resistance through the use of Ohm's law using as the voltage the difference between the voltage before the pulse and the voltage stored in the middle position of the voltage storage vector. And the current value would also be the difference between the value of the current before the pulse and the value in the middle position of the current storage vector.

With the value of the internal resistance obtained, it is enough to implement (2) and get the measurement of the battery SoH.

IV. EXPERIMENTAL RESULTS

In order to verify the efficacy of the methods to obtain the SoC and SoH, a prototype was developed. The sampling frequency is 5 kHz to estimate the SoC and 500 kHz to estimate the internal resistance.

Figure 7 presents the response of the relay for the pulse to turn it on. The delay is approximately 40 us. This delay does not affect the algorithm designed to estimate the SoC and SoH.

Figure 8 presents a zoom of the battery voltage and current when the pulse is applied. These waveforms are sent to the microcontroller through the signal conditioning circuit. The microcontroller runs the algorithm showed in Figure 6. Therefore, the internal resistance is obtained.

Figure 9 presents the relationship between the internal resistance and the number of cycles of the battery. This result was collected performing 10 cycles in the battery.

In order to verify the algorithm for the SoC estimation, the voltage battery is monitored when the battery is being charged and discharged.

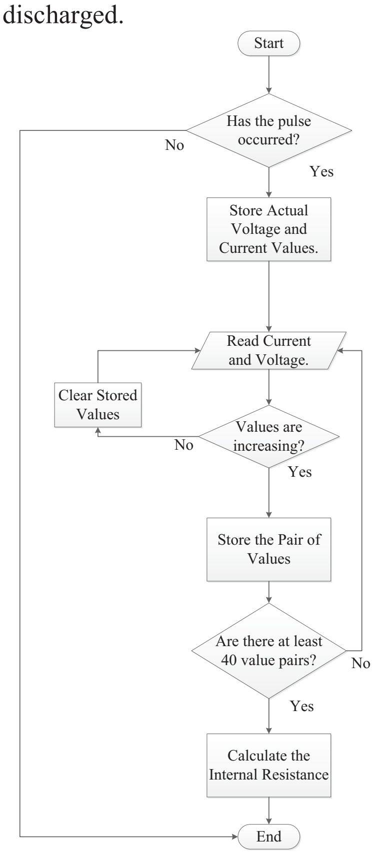

Fig. 6. Flowchart to measure the internal resistance of the battery.

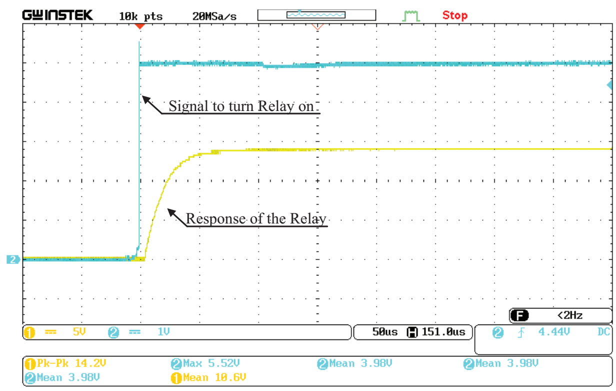

Fig. 7. Response of the relay for the pulse to turn it on.

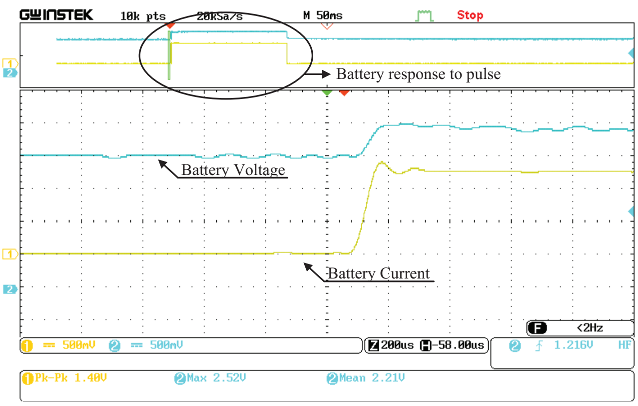

Fig. 8. A Zoom of the Battery voltage and current when the pulse is applied.

Figure 10 presents the SoC versus the battery terminal voltage when the battery is discharging. To perform this discharge was used a resistive circuit like that of Figure 4.

The resistance value was kept constant throughout the discharge.

The Figure 11 shows the voltage at the terminals and the actual current that is passing through the battery while it is being charged. The circuit used to charge the battery is shown in Figure 2. A 15V source was used to charge the 12V battery. These curves correspond to the behavior of the SoC of lead-acid batteries. Therefore, the method used to obtain the SoC is positively verified.

Figure 12 presents a picture of the prototype.

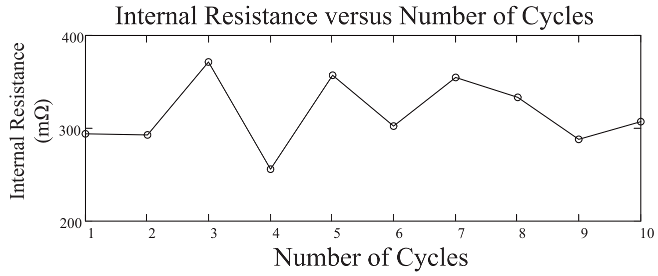

Fig. 9. Relationship between the internal resistance and the number of cycles of the battery.

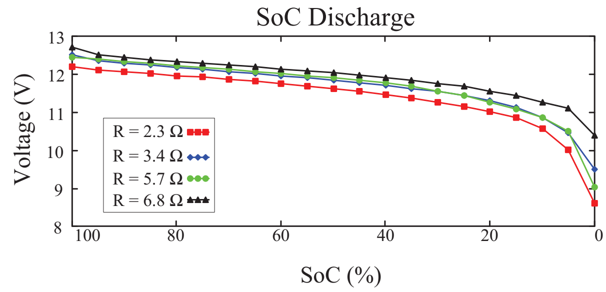

Fig. 10. SoC versus the battery terminal voltage when the battery is discharging using different resistors.

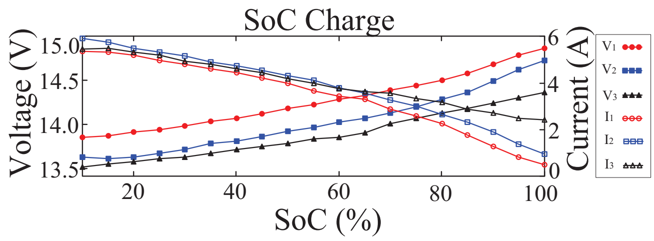

Fig. 11. SoC versus the battery terminal voltage and actual current when the battery is charging.

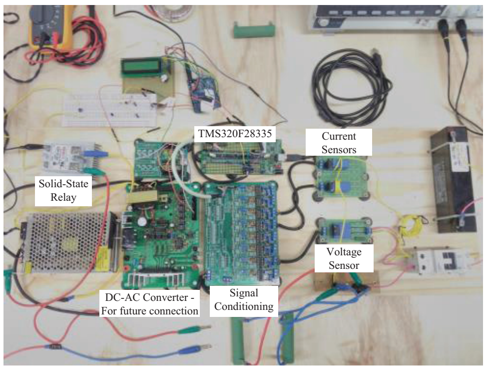

Fig. 12. The prototype.

V. CONCLUSIONS

This paper presented the design and implementation of lead-acid battery SoC and SoH. The main goal was to describe carefully the procedure to design and implement the SoC and SoH estimators. The SoH is estimated through generating a pulse to the battery and monitoring its voltage and current. Then, the SoH is obtained by calculating the battery internal resistance. On the other hand, the SoC was estimated through the Coulomb Counting method. Experimental results showed the efficacy of the proposed estimation method. The proposed estimator can be applied in real-time applications. The SoH and SoC estimators are independent of the DC/AC converter used to process the battery energy. When there is current through the battery, independently of its direction, the SoC estimators keeps measuring and computing the value of the charge within the battery. On the other hand, the proposed estimator computes the SoH value in moments when the battery begins to be charged. Therefore, the design and implementation of SoC and SoH proposed in this paper is an attractive solution for lead-acid battery applications.

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